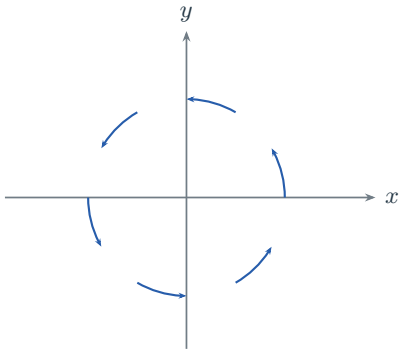
例：用直接计算核对叉乘场的散度



$\mathbf{A} = (-y, x, 0)$，$\nabla \times \mathbf{A} = 2\mathbf{e}_z$

$$\mathbf{B} = (0, 0, z),\ \nabla \times \mathbf{B} = \mathbf{0}$$
$$\mathbf{A} \times \mathbf{B} = (xz, yz, 0)$$

直接求导：

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = z + z = 2z$$

恒等式：

$$\mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) = 2z$$